

SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

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Problem Statement

- Inefficient manual tracking of student admissions and staff assignments
- Tedious and error-prone manual computation of CBC grades and report cards
- Lack of centralized, real-time daily attendance tracking across classes
- Difficulty in managing, tracking, and reconciling fee invoices and payments
- Poor communication channels for school announcements and updates
- Complex and disjointed manual scheduling for weekly school timetables

Project Objectives

System Objectives

- Centralized cloud platform for all school data
- Automated CBC grading and report card generation
- Digital attendance tracking with statistics
- Fee management with payment tracking
- Timetable scheduling and management
- Communication tools (announcementss)
- Role-based access control (Admin, Teacher, etc.)

Research Objectives

- Investigate data loss in current manual systems
- Evaluate digital literacy of stakeholders
- Design scalable CBC-aligned data model
- Implement and test full-stack web application

Technology Stack

Layer	Technology	Purpose
Frontend	React 18 + Vite + Tailwind CSS	Single Page Application UI
Backend	Node.js 20 + Express.js	RESTful API Server
Database	PostgreSQL 15 + Prisma ORM	Relational Data Storage
Auth	JWT + bcrypt (12 rounds)	Authentication & Security
State	Zustand	Frontend State Management
Hosting	Cloudflare Pages + Render + Supabase	Cloud Deployment

System Architecture

- Three-Tier Client-Server Architecture
- Client Tier: React 18 SPA served via Cloudflare Pages
- Server Tier: Express.js API with 9-layer middleware pipeline
- Database Tier: PostgreSQL with 23+ tables, UUID keys, indexed columns
- Middleware: CORS → Helmet → Rate Limit → Auth → RBAC → Validate
- Auto-deploy via GitHub to Render (backend) and Cloudflare (frontend)

Database Design

- 23+ tables designed with Prisma ORM on PostgreSQL
- UUID-based primary keys for all entities
- Key entities: User, Student, Staff, Class, Grade, Subject, Attendance
- Assessment, AssessmentScore, ReportCard, FeeStructure, Invoice, Payment
- Unique constraints prevent duplicate records (e.g., attendance per day)
- Indexed columns for fast lookups: email, admission number, dates
- 6 enums: Role, AdmissionStatus, AttendanceStatus, CompetencyRating, PaymentMethod, PaymentStatus

Modules Implemented (11 Modules — 100% Complete)

-  Authentication & RBAC (6 roles)
-  Student Management (CRUD + admissions)
-  Staff Management (assignments)
-  Academic Structure (CBC grades, terms)
-  Attendance Tracking (daily marking)
-  Assessment & Grading (CBC + traditional)
-  Report Cards (rankings, averages)
-  Fees Management (invoices, payments)
-  Communication (announcements, DMs)
-  Timetable (weekly grid)
-  Dashboards (Admin + Student)
-  Deployed to Production

Security Features

- JWT authentication with 15-minute access tokens + 7-day refresh tokens
- Password hashing with bcrypt (12 salt rounds)
- Role-Based Access Control (RBAC) middleware on every route
- Redis-backed rate limiting to prevent brute-force attacks
- Helmet.js for XSS, CSRF, and clickjacking protection
- Input validation (express-validator) + sanitization middleware
- Comprehensive audit log (action, entity, user, IP, timestamp)

Frontend — 17 Pages Implemented

Page	Description	Access
Login	Email/password authentication	All
Dashboard	Role-specific analytics & widgets	All
Students	Student list with search, filter, CRUD	Admin/Teacher
Admissions	Approval/rejection workflow	Admin
Staff	Staff management & assignments	Admin
Attendance	Daily marking interface	Teacher
Assessments	Score entry with auto-grading	Teacher
Reports	Report card generation	Admin/Teacher
Fees	Invoices, payments, balances	Bursar
Timetable	Weekly grid view	All

Testing Summary

Test Type	Tool	Tests	Pass Rate
Unit Testing	Vitest	Key service functions	100%
Integration Testing	Postman	All API endpoints	100%
User Acceptance	Manual	End-to-end workflows	100%
Cross-Browser	Manual	Chrome, Firefox, Edge, Safari	100%
Responsive	DevTools	320px – 1920px viewports	100%
Security	Manual + Helmet	Auth, RBAC, rate limiting	100%
Total		46 test cases	100%

Challenges & Solutions

Challenge	Solution
Supabase connection failures	Connection pooling + Prisma retry logic
Data loss after page refresh	Removed auto-seeding from login flow
Assessments not grouped by term	Refactored query with term relations
Dashboard showing placeholder data	Replaced with live API calls
Rate limiting in production	Redis-backed store with graceful fallback



LIVE DEMO

Future Enhancements

- M-Pesa STK Push Integration — direct mobile money payments
- SMS / Email Notifications — automated alerts for attendance & fees
- Flutter Parent App — native mobile app for parents
- PDF Report Cards — server-side PDF generation with PDFKit
- Multi-School Support — extend architecture for multiple institutions

Conclusion

- All 11 modules successfully designed, developed, tested, and deployed
- 100% functional requirement coverage achieved
- System addresses: manual records, data loss, slow reports, poor communication
- Fully compliant with Kenya's CBC grading framework
- User acceptance testing completed with school administrators
- System is live and accessible to all stakeholders

THANK YOU

Questions?

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GitHub: <https://github.com/MuthamiM/KYAMATU-SMS>